ABSTRACT

Anviksha -8 Bit project presents the design, simulation, and hardware implementation of a basic 8-bit computer system developed using the Logisim Evolution. The primary objective is to create an educational processor architecture that demonstrates the fundamental principles of CPU operation at the hardware level while bridging the gap between theoretical concepts and practical implementation.

The proposed architecture integrates essential computational components, including an Arithmetic Logic Unit (ALU), registers, a program counter, a control unit, an instruction decoder, and Static Random-Access Memory (SRAM). The processor adopts a microcoded control architecture and implements a Complex Instruction Set Computer (CISC) instruction set capable of executing arithmetic operations, data transfer, conditional branching, and program control instructions.

To extend the design beyond simulation, the processor is implemented on the Basys 3 Field-Programmable Gate Array (FPGA) development board, powered by the Xilinx Artix-7 FPGA. Hardware development and verification are carried out using Verilog Hardware Description Language (HDL) and Xilinx Vivado, enabling real-time execution, debugging, and validation of the processor design.

The functionality of the system is verified through the successful execution of sample programs, including multiplication and Fibonacci sequence generation. By prioritizing architectural clarity over modern high-performance complexity, this project provides a practical platform for understanding processor design, digital logic, and computer architecture, making it a valuable educational resource for students and researchers.